STUDY PROGRAMS

I. ID

Subject Code:	IF30020
Subject Title:	Operating Systems
Prerequisite :	None

Year:	3rd	Semester:	1st	Number of sessions:	2

Total workload (60-minute clock hours)	85	Hours completed	85		Independent work hours	17
			Theoretical Hours	Practical Hours		
			51	34		
Academic credits	6.7	Credits for hours completed	5.6		Credits for self-employed hours	1.1

II. Rationale:

Operating systems are an essential part of any computer system. Similarly, a course on operating systems is an essential part of any computer science education. This field is undergoing rapid change, as computers are now prevalent in virtually every aspect of daily life, from devices embedded in automobiles to the most sophisticated planning tools for governments and multinational corporations. This introductory course on operating systems provides a clear overview of the concepts underlying them.

III. Course purpose:

The aim of this course is to provide an introduction to the inner workings of modern operating systems. In particular, the course will cover processes and threads, mutual exclusion, CPU scheduling, deadlock, memory management, and file systems.

IV. Skills and abilities to be developed

After successfully completing the course, students will be able to demonstrate knowledge and understanding of:

- The main functionality of modern operating systems: processes/threads, scheduling, virtual memory and file systems, aspects of parallelism, kernels, shells , microkernels , virtual machines.
- Key concepts and algorithms in operating system implementations: synchronization, deadlock prevention/prevention, memory management, processor scheduling, disk scheduling, virtual machines, file system organization.
- Implementation of simple OS components.

V. Contents: (detail Units and Topics to be developed)

Unit 1 Introduction

1.1 What do operating systems do?
1.2 Organization of the computer system
1.3 Computer system architecture
1.4 Operating system structure
1.5 Operating system operations
1.6 Process Management
1.7 Memory Management
1.8 Storage Management
1.9 Protection and security
1.10 Kernel Data Structures
1.11 Computer environments

1.12 Open source operating systems

Unit 2. Operating System Structures

2.1 Operating system services
2.2 User interface and operating system
2.3 System Calls
2.4 Types of system calls
2.5 System Programs
2.6 Operating system design and implementation
2.7 Operating system structure
2.8 Operating system debugging
2.9 Operating system generation
2.10 System Startup

Unit 3. Processes

3.1 Process Concept
3.2 Process Programming
3.3 Operations on processes
3.4 Inter-process communication
3.5 Examples of IPC systems
3.6 Communication in client-server systems

Unit 4. Threads

4.1 Summary
4.2 Multicore Programming
4.3 Multi-threaded models
4.4 Thread Libraries
4.5 Implicit threading
4.6 Subprocessing Problems
4.7 Examples of operating systems

Unit 5. Process Synchronization

5.1 Background
5.2 The critical section problem
5.3 Peterson's Solution
5.4 Synchronization Hardware
Mutex Locks
5.6 Traffic lights

5.7 Classic Synchronization Problems
5.8 Monitors
5.9 Synchronization Examples
5.10 Alternative approaches

Unit 6. CPU Programming
6.1 Basic concepts
6.2 Programming criteria
6.3 Algorithm Programming
6.4 Subprocess Scheduling
6.5 Multiprocessor Programming
6.6 Real-time CPU scheduling
6.7 Examples of operating systems
6.8 Algorithm Evaluation

Unit 7. Interlock
7.1 System Model
7.2 Interlock Characterization
7.3 Methods for handling deadlocks
7.4 Interlock Prevention
7.5 Interlock Avoidance
7.6 Interlock detection
7.7 Recovering from a standstill

Unit 8. Main Memory
8.1 Background
8.2 Exchange
8.3 Contiguous memory allocation
8.4 Segmentation
8.5 Pagination
8.6 Page Table Structure
8.7 Example: Intel 32-bit and 64-bit Architecture
8.8 Example: ARM Architecture

Unit 9. Virtual Memory
9.1 Background
9.2 Pagination on demand
9.3 Copy in writing

9.4 Page Replacement
9.5 Frame Allocation
9.6 Blows
9.7 Files allocated in memory
9.8 Kernel Memory Allocation
9.9 Other considerations
9.10 Examples of operating systems
Unit 10. Mass storage structure
10.1 Overview of the mass storage structure
10.2 Disk Structure
10.3 Disk Attachment
10.4 Disk Programming
10.5 Disk Management
10.6 Exchange space management
10.7 RAID Structure
10.8 Stable Storage Implementation
Unit 11. File System Interface
11.1 File Concept
11.2 Access methods
11.3 Disk and directory structure
11.4 Mounting the file system
11.5 Sharing files
11.6 Protection
Unit 12. File System Implementation
12.1 File system structure
12.2 File system implementation
12.3 Directory Implementation
12.4 Allocation methods
12.5 Free space management
12.6 Efficiency and performance
12.7 Recovery
12.8 NFS
12.9 Example: The WAFL File System
Unit 13. I/O Systems

13.1 Summary
13.2 I/O Hardware
13.3 Application I/O Interface
13.4 Core I/O Subsystem
13.5 Transforming I/O requests into hardware operations
13.6 Sequences
13.7 Performance

Unit 14. Protection

14.1 Protection objectives
14.2 Protection Principles
14.3 Protection Domain
14.4 Access Matrix
14.5 Implementation of the access matrix
14.6 Access control
14.7 Revocation of access rights
14.8 Capacity-based systems
14.9 Language-based protection

Unit 15. Security

15.1 The security problem
15.2 Program Threats
15.3 System and network threats
15.4 Cryptography as a security tool
15.5 User Authentication
15.6 Implementation of security defenses
15.7 Firewalls to protect systems and networks
15.8 Computer security classifications
15.9 An example: Windows 7

VI. Teaching strategies for the teaching-learning process:

- Theoretical classes, accompanied by practical laboratory classes.
- Practical work.

VII. Evaluation strategies:

- Process: 20%
- Partial test: 40%
- Final exam: 40%

VIII. Bibliographic sources: (minimum five basic bibliographies and three supplementary ones)

Basics

- Silberschatz , A. (2012). Operating System Concepts. John Wiley & Sons , Inc.
- Deitel , H. M. (1983). An Introduction to Operating Systems. Addison-Wesley Publishing Company.
- Stallings, W. (2001). Operating Systems: Internals and Design Principles. Prentice Hall.
- Tanenbaum, A.S. (2005). Operating Systems Design and Implementation.
- Tanenbaum, A.S. (2014). Modern Operating Systems. Prentice Hall Press .

Additional information:

- https://youtube.com/playlist?list=PLBlnK6fEyqRiVhbXDGLXDk_OQAeuVcp2O
- https://youtu.be/vLPdBp9vv9Y
- https://edu.gcfglobal.org/en/computerbasics/understanding-operating-systems/1/